

CSA431 · COMPUTER VISION · LITERATURE REVIEW



Patient- vs Cell-Level Evaluation in Malaria Parasite Classification

Dataset: NIH / NLM Malaria Cell Images · 27,558 cells

Base paper: Rajaraman et al., 2018 — PeerJ 6:e4568

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MOTIVATION

Why evaluation protocol matters here

27,558

total cell images
(13,779 + 13,779)

150 / 50

infected vs healthy
patients sampled

~200

source slides —
cells are not independent

- Screening today still relies on manual microscope reading of Giemsa-stained blood smears.
- Many later papers report 96–99% accuracy on this exact dataset — the number keeps climbing.
- Course rule for this project: report sensitivity/recall + AUC, not accuracy alone; no clinical claims.

Random split vs. patient-disjoint split



Random cell-level split

Cells are shuffled and split without checking their source slide.

Cells from the SAME slide can land in both train and test — the model partly memorises stain, lighting and focus, not just the parasite.



Patient- / slide-disjoint split

Every slide is assigned fully to train OR test — never both.

A test slide is completely unseen. This is the fair way to check if the model generalises to a new patient.

Rajaraman (2018) already tested both: **98.6% cell-level** → **95.9% patient-level** (Table 6). They noted that, at the time, no comparable work had run patient-level CV at this scale — but later NIH papers largely did not repeat that protocol.

Papers that actually compared both splits

Paper	Method	Split	Acc	Sens	AUC
Rajaraman 2018	VGG-16 / ResNet-50 features	BOTH: cell-level AND patient 5-fold	0.986 / 0.959	0.981 / 0.947	0.999 / 0.991
MalariaNet 2026	Compact CNN, 8 architectures	BOTH: per-cell 70/15/15 AND slide-disjoint	0.971 / 0.956	0.939 (slide-disj.)	0.988 (slide-disj.)

TAKEAWAY

- ✓ Leakage is real — the drop shows up consistently across all 8 architectures in MalariaNet.
- ✓ But the drop is small: roughly 1.5 percentage points, inside cross-seed noise (± 1.02).
- ✓ So the direction of the bias is the finding here — not a catastrophic accuracy collapse.

High accuracy, weaker split reporting

Paper	Method	Split we could verify	Headline acc
Narayanan 2019	GoogLeNet / ResNet + CAM	80/20 hold-out, cell-level; 10% of train = val	0.965–0.966 · AUC 0.993
Alharbi 2022	VGG-19 fine-tune	Cell-level random; not patient-aware	0.9834 ± 0.0051
EDRI 2024	EfficientNetB2 + Dense/Res/Inception	Stratified 80/10/10; no k-fold	0.9768 · recall 0.9644
Mmileng 2025	ConvNeXt V2 Tiny Remod	Split ratio not stated; heavy augmentation	0.981 (V2) · 0.959 (V1)
ISTNet 2026	Inception–Swin hybrid	Not in abstract; full text unavailable	0.9661 NIH · 0.9965 BBBC041
CNN-ViT 2025	ResNet18 + ViT_B_16 ensemble	80/20 + k-fold claimed; not patient-disjoint	0.9964 · recall 0.9975

Also (meta / labels, not accuracy papers): PRISMA review 2024 (field-level corroboration, full text not retrieved) · Wilm et al. 2025 — COCO boxes on NIH smears, border cells unlabeled, infected-cell F1 up to 0.88.

What the table actually shows

2 / 10

papers ran a leakage-free comparison — Rajaraman and MalariaNet

Several

high-accuracy papers use random or stratified cell splits, or never state a patient split at all

95.9%

ConvNeXt V1 Tiny matches Rajaraman's patient-level number — but for an unrelated reason

Cross-dataset $\neq$ fix

testing NIH + BBBC041 (ISTNet) is not a substitute for a within-dataset patient split

CNN-ViT (2025) has the most complete metric list in the whole survey — yet still no patient-disjoint protocol.

The gap we are building on

1

Random cell splits leak slide identity — Rajaraman already measured 98.6% → 95.9% under patient-level CV, and later NIH papers largely did not repeat that protocol.

2

MalariaNet (2026) confirms the leak across ~200 slides but shows the drop is only ~1–2 pp and sits inside cross-seed noise — so "leakage wrecks accuracy" is too simple.

3

Mean accuracy hides the real risk: a few hard slides can dominate error, and sensitivity vs. specificity can move differently than accuracy.

4

Open question we study: under slide-disjoint evaluation, how does error vary per slide — and does sensitivity drop more than specificity?

One change, rubric-aligned

Base

Rajaraman 2018 ResNet-50 feature extractor, NIH 27,558 cells

Keep

Same architecture, same patient/slide folds as far as filenames allow

Change

Stop reporting only mean accuracy — add (1) per-slide error distribution (2) sensitivity vs. specificity under the same leakage-free split

Predict before coding

Mean-accuracy drop will be small (~1–2 pp, matching MalariaNet); per-slide variance and sensitivity will move more than accuracy

Metrics we always report: sensitivity/recall, specificity, AUC — never accuracy alone. *Not a diagnostic claim — screening experiment on public cells only.*

Why this base paper

Why Rajaraman 2018



Public dataset released by the authors; code-reproducible CNN feature-extraction setup



Only origin paper publishing both cell-level and patient-level numbers on this dataset

Reproducibility checklist



Dataset URL public · 5-fold patient CV described · metrics listed



Limitations stated: staining variation across patients

Note: MalariaNet is the 2026 confirmation paper — not the base we reproduce.

Where this leaves us

- 1 The field still publishes 97–99% on NIH with cell-level or unstated splits.
- 2 Leakage exists; the interesting leftover gap is metric- and slide-level, not a huge accuracy gap.
- 3 Next: reproduce the Rajaraman patient-level baseline, then run the one analysis change above.

LIMITATIONS WE ALREADY EXPECT

Single-species thin-smear cells · Label noise at smear borders (Wilm 2025) · No external hospital set in M1